

Mansi Budamagunta

Personal Information

Email b.mansi12@gmail.com
Phone +91-7066021768
Citizenship Indian

Education

2025 – present **Utrecht University**
MS Applied Data Science, GPA (so far): 8.9

2016 – 2021 **Indian Institute of Science Education and Research (IISER) Pune**
BS-MS Dual Degree, Interdisciplinary Major, GPA: 8.9

Selected coursework

- **Mathematics:** Linear Algebra, Probability and Statistics, Calculus, Numerical Analysis, Graph Theory, Algorithms, Operations Research, Combinatorics, Mathematical Methods in Physics
- **Other:** Statistical Mechanics, Mathematical & Computational Biology, Neurobiology, Ecology, Law Science and Society, Introduction to Development Studies

Publications

- 2022 **Mansi Budamagunta**, Anmol Gaur, Ashutosh Verma, Arunim Anand, and Chirag Deb. "Clustering Indian Districts Based on Multidimensional Demographic and Climate Data." In *Proceedings of the 9th ACM International Conference on Systems for Energy-Efficient Buildings, Cities, and Transportation*, 450-455. New York: ACM, 2022. <https://doi.org/10.1145/3563357.3566146>.
- 2021 Onkar Sadekar, **Mansi Budamagunta**, G. J. Sreejith, Sachin Jain, and M. S. Santhanam. "An Infectious Diseases Hazard Map for India Based on Mobility and Transportation Networks." *Current Science* 121, no. 9 (2021): 1208-1215. <https://doi.org/10.18520/cs/v121/i9/1208-1215>.

Industry Experience

January 2023 – March 2025 **AI Trainer at Turing**
Worked as a Python programmer as part of a global team at Turing (a company that provides custom data solutions to top AI labs like OpenAI, Anthropic, Google, Meta, etc. for the development of their foundational models). Analyzed LLM outputs to enhance coding and reasoning capabilities in the RLHF feedback loop. Contributed to the development of several tools integrated with modern AI systems like plugins, browsing, coding, and computer usage. Led teams of varying sizes for a year, optimizing workflows and establishing quality standards.

Research Experience

- July 2022 – January 2023 **Urban Data Analytics - Clustering Indian Cities**
Advisor: Prof. Chirag Deb (IIT Bombay)
Clustered Indian Urban Districts using remote sensing, climate, census and demographic data. Collected and collated these diverse datasets for 96 districts, performed k-means clustering to identify patterns, and revealed that geographic proximity doesn't always determine similarity of districts. Presented this research at BuildSys '22: Proceedings of the 9th ACM International Conference on Systems for Energy-Efficient Buildings, Cities, and Transportation ([DOI](#)).
- June 2020 – June 2022 **Spreading Processes on Transportation Networks**
Advisor: Prof. M. S. Santhanam (IISER Pune)
Constructed networks based on Indian census data for 446 major cities and the rail & air transport networks connecting them. Modeled infectious disease spread through these transportation networks using SIR metapopulation models. You can find more information at [the project website](#) or the resulting research article, [Current Science 121, 1208 \(2021\)](#).
- May 2018 – March 2020 **Conductance based modelling of neuronal networks to study brain rhythms**
Advisor: Prof. Collins Assisi (IISER Pune)
Studied theta and gamma rhythms in the medial Entorhinal Cortex (mEC) of rodent brains using networks of conductance-based models of neurons.
- May 2019 – July 2019 **Connectome based whole brain modelling**
Advisor: Prof. Viktor Jirsa (Aix-Marseille Université)
Analyzed connectome-based whole-brain models via numerical simulations to study the structure-function relationship in human and rodent brains. Also contributed to the development of the software *The Virtual Brain* in association with the European Human Brain Project.

Skills

Data Science: Machine Learning, Data Analysis, Network Analysis, Data Visualization

Programming: Proficient in Python; Comfortable with HTML, CSS, Javascript; Familiar with R, C, C++

Software: Git, Jupyter, Docker, Linux terminal

Languages: Fluent in English, Hindi, Telugu; Beginner French

Honors and Scholarships

- 2016 - 2021 INSPIRE scholarship holder - Awarded by Department of Science and Technology (DST), Government of India for consistently strong academic performance.

Extra-curricular activities

- June 2017 Part of a nine-member Indian Youth Delegation to Vietnam selected by the Ministry of Youth Affairs & Sports.
- August 2016 – July 2018 Taught underprivileged children at a nearby slum as a regular volunteer and then program coordinator of one of the programs of Disha, the social outreach club of IISER-Pune.
- February 2017 – February 2020 Participated in annual science outreach activities in villages near Pune as part of one of the programs of Disha, the social outreach club of IISER-Pune.